Yiwei Shu

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EDUCATION

University of Chinese Academy of Sciences, Beijing, China — Sept 2024 – present
Ph.D. in Computer Science and Technology

- Technology and Engineering Center for Space Utilization, Chinese Academy of Sciences.
- Research focus on optimization-based navigation and autonomous exploration for legged robots in complex and planetary environments, with emphasis on risk-aware planning, multi-gait decision-making, and active perception.

ShanghaiTech University, Shanghai, China — Sept 2020 – June 2024
Bachelor of Engineering in Electronic Information Engineering

University of Illinois Urbana-Champaign, IL, United States — Dec 2022 – May 2023
Exchange Student; earned grades of A or A+ in all courses.

RESEARCH EXPERIENCE

Optimization-Based Multi-Gait Navigation for Quadruped Robots in Lunar Environments — July 2026 – present
Technology and Engineering Center for Space Utilization, Chinese Academy of Sciences

- Develop a goal-directed navigation framework that jointly coordinates autonomous exploration and adaptive locomotion for time-efficient traversal of GPS-denied lunar terrain.
- Integrate LiDAR-inertial state estimation, occupancy-grid mapping, active perception, and risk-aware point-goal planning on the Unitree Go2.
- Formulate gait-conditioned planning models that jointly optimize traversal time, stability, energy consumption, and terrain risk.
- Build reduced-gravity simulation benchmarks in Isaac Lab for multi-gait policy evaluation, uncertainty analysis, and sim-to-real transfer.

SELECTED MANUSCRIPTS

From End-Effector Targets to Whole-Arm Postures: Configuration-Space Waypoint Recovery for Reactive Manipulation — 2026
Yiwei Shu, Shuquan Wang. *Under review at IEEE Robotics and Automation Letters*.

- Developed PostureWaypoint, which lifts Cartesian bypass targets into collision-free, joint-feasible whole-arm configurations, resolving posture ambiguity without global replanning or task-specific training.
- Formulated an event-triggered pipeline combining geometric SDF-based bypass generation, SQP/Gauss-Newton posture synthesis, posture-guided reactive control, and a progress-preserving CBF-QP safety filter; the complete framework runs at 50 Hz with 2.73 ms average computation per step.
- Evaluated on four manipulators and six constrained 3D tasks across 1,200 randomized episodes per method, achieving the highest mean whole-arm success rate of 70.2%, 5.5 percentage points above the strongest baseline. RM75 hardware tests achieved 20/20 success in both real-robot tasks.

Finite-State Visual Servo Control for Autonomous Target Landing of Low-Cost Underactuated Fixed-Wing Aircraft — 2026
Yiwei Shu, Shuquan Wang. *Under review at Aerospace Science and Technology*.

- Developed a finite-state controller that decomposes short-duration unpowered landing into IMU-guided ascent and monocular vision-guided descent, enabling target landing under limited sensing and weak control authority.
- Independently designed and integrated a 189 g aircraft, onboard electronics, control-surface actuation, and an adjustable catapult launcher; the complete platform had a total hardware cost of CNY 534 and required no GPS, airspeed sensor, or high-grade inertial navigation system.
- Established a CFD-informed simulation-to-flight pipeline with randomized disturbances, noise, delay, and aerodynamic uncertainty; reduced the normalized final image-plane radial error from 0.5049 to 0.0027 and the RMS error from 1.4526 to 0.1732 in simulation, while achieving 20/20 real-world landings.

RKILPlanner: Risk-Based Kinematic Imitation Learning Planning Towards Human-Like Driving in Highway Scenarios 2026

Yiwei Shu, Shuquan Wang, Jun Li. *Submitted to ROBIO 2026*.

- Developed a risk-aware imitation-learning planner that integrates Driver's Risk Fields, personalized driving styles, and differentiable vehicle kinematics for physically feasible highway trajectory generation.
- Evaluated on 28,641 HighD samples, achieving 0.159 m ADE, 0.443 m FDE, and a 0.3% miss rate over a safety-critical 2.5 s planning horizon.

PUBLICATIONS

Expanding Grasp Reachability: Field-Based Pose-Constrained Two-Stage Grasp Planning 2026

Yiwei Shu, Shuquan Wang, Jun Li, Hanying Sang, Yidong Ye. *ICoSR 2026, accepted full paper and oral presentation*.

- Developed a field-based two-stage grasp planner that uses reachability and pose-aware manipulability models to relocate difficult targets into more dexterous regions before pickup.
- Validated on a 7-DoF RM75 in simulation and hardware, achieving 27/30 boundary grasps versus 0/30 for direct picking and expanding the real-world pose-constrained grasp radius from 0.38 m to 0.57 m.

Learning-Based Extended-Workspace Manipulation: A Self-Relocating Space Crawler for Autonomous Deep-Space Outposts 2026

Yiwei Shu, Shuquan Wang. *46th COSPAR Scientific Assembly, poster presentation*.

- Developed a MuJoCo-based self-relocating space-crawler framework that alternates anchor selection, inverse-kinematics-driven grasping, base translation, and target manipulation to extend the reachable workspace of an orbital servicing robot.
- Investigated learning-based anchor selection for autonomous relocation and manipulation in deep-space assembly and outpost construction scenarios.

Attitude Control of Unpowered Aircraft in Flight Based on Kalman Filter Visual Servo-PID Optimization Algorithm 2025

Yiwei Shu, Shuquan Wang. *44th Chinese Control Conference (CCC 2025), accepted full paper and oral presentation*.

- Developed a Kalman-filtered visual servo-PID controller for low-cost unpowered aircraft, enabling robust attitude regulation and trajectory correction using only monocular visual feedback.
- Reduced the average trajectory error under Gaussian noise to 1.3650, outperforming sliding-mode control and standard visual servoing by 48.4% and 41.1%, respectively.

SKILLS

Programming: Python, C, C++, R, Verilog

Robotics & Simulation: ROS/ROS 2, PCL, Isaac Lab, MuJoCo, PyTorch, Unitree Go2, RM75

Engineering Tools: Linux, Git, STM32, SolidWorks, Altium Designer, XFlow, Cadence, Vivado

LEADERSHIP AND SERVICE

UCAS Student Union 2024 – present

Department Head, Student Life and Culture

- Lead department operations and coordinate university-wide cultural programs, student-life initiatives, and major campus events.

ShanghaiTech RoboMaster Team 2021 – 2024

Sentry Team Lead

- Coordinated embedded programming, PCB design, mechanical modeling, and team planning for the autonomous sentry robot.

ShanghaiTech University

Teaching Assistant, Digital Circuit Course

- Led tutorials, answered technical questions, and assisted with assignment and examination grading.

ShanghaiTech University Social Practice Program 2021

Volunteer Teacher

- Taught at a local primary school and conducted field research on local education conditions.

HONORS AND AWARDS

- Outstanding Student Award, University of Chinese Academy of Sciences (2025-2026)
- Excellence Rising Star Award, UCAS Student Union (2025-2026; 2024-2025)
- Outstanding Student Leader Award, University of Chinese Academy of Sciences (2024-2025)
- The 6th National Undergraduate IC Innovation and Entrepreneurship Competition Second Prize in the East China Competition Area (2022)
- RoboMaster Competition, National 3rd Award and Eastern Division 2nd Award (2022)